

Predicting High-Unusual Coastal Debris Events Using Machine Learning

1. Abstract

This project uses machine learning to predict coastal cleanup events with unusually high shares of nonstandard debris. Using cleanup records from 2015–2026, I constructed a supervised binary classification task in which each cleanup event was labeled as either a normal debris event or a high-unusual-debris event. The target variable was based on `unusual_share`, defined as the proportion of unusual debris items within all collected items.

I first trained models on the full 2015–2026 dataset. Gradient Boosting performed best, achieving a ROC-AUC of about 0.842 and a PR-AUC of about 0.638. However, permutation feature importance showed that `year` was the most important predictor, raising concern that the model may have learned changes in reporting structure or item categories across years rather than only real debris patterns. To address this issue, I performed a no-year robustness check and then built a cleaner 2023–2025 model using years with more consistent item category structures.

In the 2023–2025 clean model, Gradient Boosting again performed best, with a ROC-AUC of about 0.752 and a PR-AUC of about 0.547. Feature importance showed that geographic variables, especially `zone_name`, longitude, and latitude, were the strongest predictors. Overall, the results suggest that high-unusual-debris events are spatially uneven and can be predicted using cleanup-level information, but careful interpretation is necessary because data structure and reporting practices strongly affect model behavior.

2. Introduction

Coastal cleanup data usually contain many common trash items, such as cigarette butts, plastic bottles, bottle caps, food wrappers, and plastic bags. However, some cleanup events contain a high share of less common or nonstandard debris, such as fishing gear, tires, construction materials, household items, electronic waste, hazardous containers, or other large objects. These high-unusual-debris events may reflect local environmental conditions, cleanup location, human activity, storm impacts, or differences in reporting and cleanup practices. Being able to predict the likelihood of high-unusual-debris events can help prepare appropriate tools and enough labor forces for a specific cleanup activity.

The purpose of this project is to explore whether machine learning can help identify cleanup events that are likely to contain unusually high shares of nonstandard debris. Instead of only describing debris patterns after cleanup, this project frames the problem as a predictive modeling task: given information about the cleanup event, location, date, cleanup type, and cleanup effort, can a model predict whether the event will be classified as high-unusual-debris?

This project is also motivated by a broader interest in natural hazards and coastal resilience. Unusual debris may be connected to local infrastructure, marine activity, tourism, waste management, or disaster-related debris movement. Although this project does not prove causal relationships, it provides a first step toward using cleanup data and machine learning to identify spatial patterns in unusual coastal debris.

3. Research Questions

The main research question is:

Can cleanup information predict whether a coastal cleanup event will have a high share of unusual debris?

This main question is divided into several subquestions:

1. Which machine learning model performs best for predicting high-unusual-debris cleanup events?
2. Does the full 2015–2026 dataset contain meaningful predictive signals?
3. Does the full model rely too heavily on `year`, possibly because of changes in item category structure or reporting practices?
4. If the model is restricted to the cleaner 2023–2025 period, can it still predict high-unusual-debris events?
5. Which features are most important in the clean 2023–2025 model?

4. Data Source and Unit of Analysis

The dataset consists of coastal cleanup records downloaded as yearly CSV files. Each row represents one cleanup event. The full dataset covers cleanup events from 2015 to 2026. Hence, the unit of analysis is one cleanup event.

The full 2015–2026 dataset was used for exploratory modeling and diagnostic analysis. However, the item category structure changed across years, and 2026 is only a partial year. Therefore, a cleaner 2023–2025 dataset was also constructed for the main final model. The 2023–2025 period was selected because these years have a more consistent category structure and represent complete years without missing too many values like reports in 2015 and 2016.

The main types of variables used in this project include:

Variable Type	Examples
Time variables	<code>year</code> , <code>month</code> , <code>hurricane_season</code>
Location variables	<code>state_name</code> , <code>zone_name</code> , <code>Latitude</code> , <code>Longitude</code>

Cleanup effort variables	People, Distance km, Area km
Cleanup type	Cleanup Type
Target construction variables	Total Items Collected, unusual_debris_count, unusual_share

Variables such as **Total Items Collected**, **unusual_debris_count**, and **unusual_share** were used to construct the target variable, but they were not used as model features. Including them as features would create data leakage because they are only known after the cleanup has occurred.

Hurricane_season was assigned “1” if cleanup happened between June to November, and was assigned “0” otherwise.

5. Data Cleaning

The raw dataset required several cleaning steps before it could be used for machine learning. First, I loaded yearly CSV files from Ocean Conservancy’s cleanup report in the United State of America between 2015 to 2026 and removed total summary rows that did not represent individual cleanup events. I then standardized the columns across years because different yearly files contained different item categories. The cleaned dataset used categories of 2026 report as a consistent structure so that yearly files could be combined.

Date variables were converted into proper datetime format for analysis. The **year** and **month** variables were extracted from the cleanup date. Numeric columns such as item counts, number of people, distance cleaned, and area cleaned were converted into numeric format. Missing item count values were filled with zero because it means that the item was not recorded in that cleanup event. Rows with invalid or zero total item counts were removed because these cleanup events are not statistically significant to calculate **unusual_share**.

The final full modeling dataset contained **131,909 cleanup events**. For the clean 2023–2025 model, the dataset was restricted to complete recent years with more consistent item category definitions.

$$\text{unusual_share} = \text{unusual_debris_count} / \text{Total Items Collected}$$

Then I set the cleanup activity with the highest 25% of the unusual share as the cutoff for high unusual events. For the 2023–2025 main model, **high_unusual_event** was reconstructed within the 2023–2025 subset. This avoids using the cutoff from 2015-2026 full model that influenced that accuracy of the model result. Events in the top 25% of **unusual_share** were labeled as 1, and the rest were labeled as 0.

6. Defining Unusual Debris and Constructing the Target Variable

A key step in this project was defining what counts as “unusual debris.” I defined unusual coastal debris as **large, hazardous, household, industrial, fishing-related, or disaster-related items** that are not typically classified as everyday consumer litter. Examples include fishing gear, lines and nets, tires, construction materials, household objects, electronic waste, PPE, syringes, appliances, and other nonstandard debris types.

For each cleanup event, I calculated:

$$\text{unusual_share} = \text{unusual_debris_count} / \text{Total Items Collected}$$

Then I constructed a binary target variable:

$$\begin{aligned} \text{high_unusual_event} &= 1 \text{ if unusual_share is in the top 25\%} \\ \text{high_unusual_event} &= 0 \text{ otherwise} \end{aligned}$$

This means that the model predicts whether a cleanup event belongs to the highest quartile of unusual debris share.

For the full 2015–2026 model, the top 25% cutoff was calculated using the full modeling dataset. For the 2023–2025 clean model, the target variable was reconstructed within the 2023–2025 subset as mentioned before.

7. Exploratory Data Analysis

7.1 Target Distribution

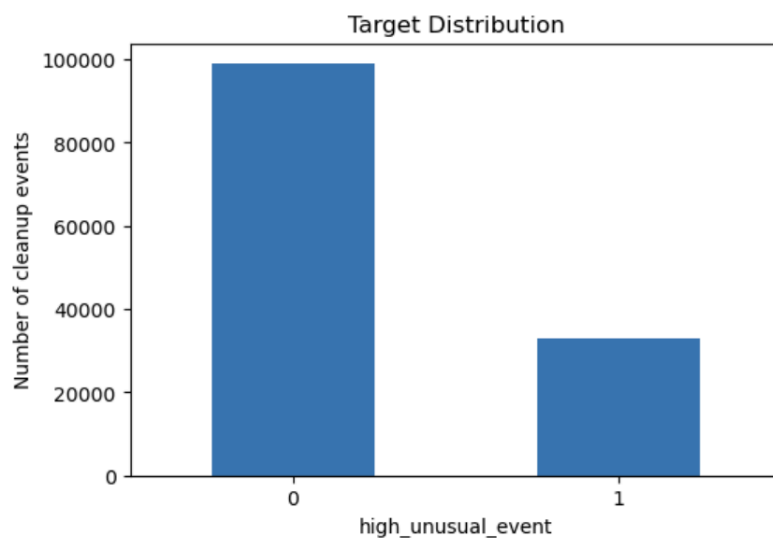
In the full 2015–2026 modeling dataset, 98,895 cleanup events were labeled as normal events, while 33,014 were labeled as high-unusual-debris events. This means that approximately 75.0% of observations are normal events and 25.0% are high-unusual events.

This distribution is expected because the target variable was constructed using the top 25% of `unusual_share`. Although this imbalance is not extreme, it still affects model evaluation. Accuracy alone is not sufficient because a model could perform well on the majority class while failing to identify high-unusual-debris events. To provide meaningful analysis, F1-score, ROC-AUC, and PR-AUC are also used to evaluate model performance.

Figure 1. Target distribution for the full 2015–2026 dataset.

```
Target counts:
high_unusual_event
0    98895
1    33014
Name: count, dtype: int64
```

```
Target proportions:
high_unusual_event
0    0.749721
1    0.250279
Name: proportion, dtype: float64
```



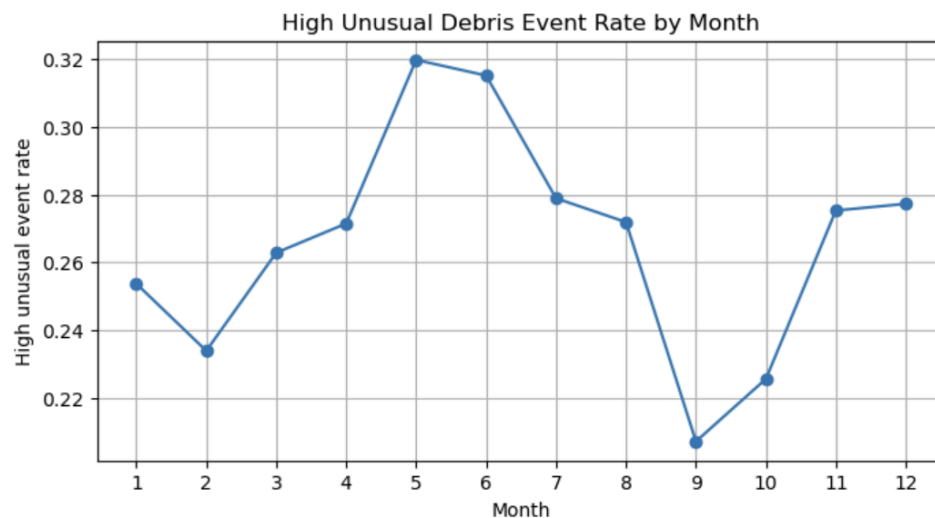
7.2 Monthly Pattern

The monthly analysis shows that high-unusual-debris event rates vary across the year. The rate is highest in May and June, with both months above 31%. September has the lowest high-unusual rate, around 20.7%, while it has the highest number of cleanup events which is 3 times more than October, the second highest number of cleanup events..

This pattern suggests that month contains useful seasonal information, but the pattern does not simply follow hurricane season. September's large number of events may reflect organized cleanup campaigns that collect a large amount of ordinary litter, which can reduce the unusual debris share. This supports including `month` as a model feature.

Figure 2. High-unusual-debris event rate by month.

	month	number_of_events	high_unusual_rate
0	1	5133	0.253653
1	2	4760	0.233824
2	3	6796	0.262802
3	4	11436	0.271424
4	5	9219	0.319774
5	6	9138	0.315167
6	7	8362	0.278881
7	8	7769	0.271850
8	9	43503	0.207043
9	10	13653	0.225518
10	11	7498	0.275273
11	12	4642	0.277251

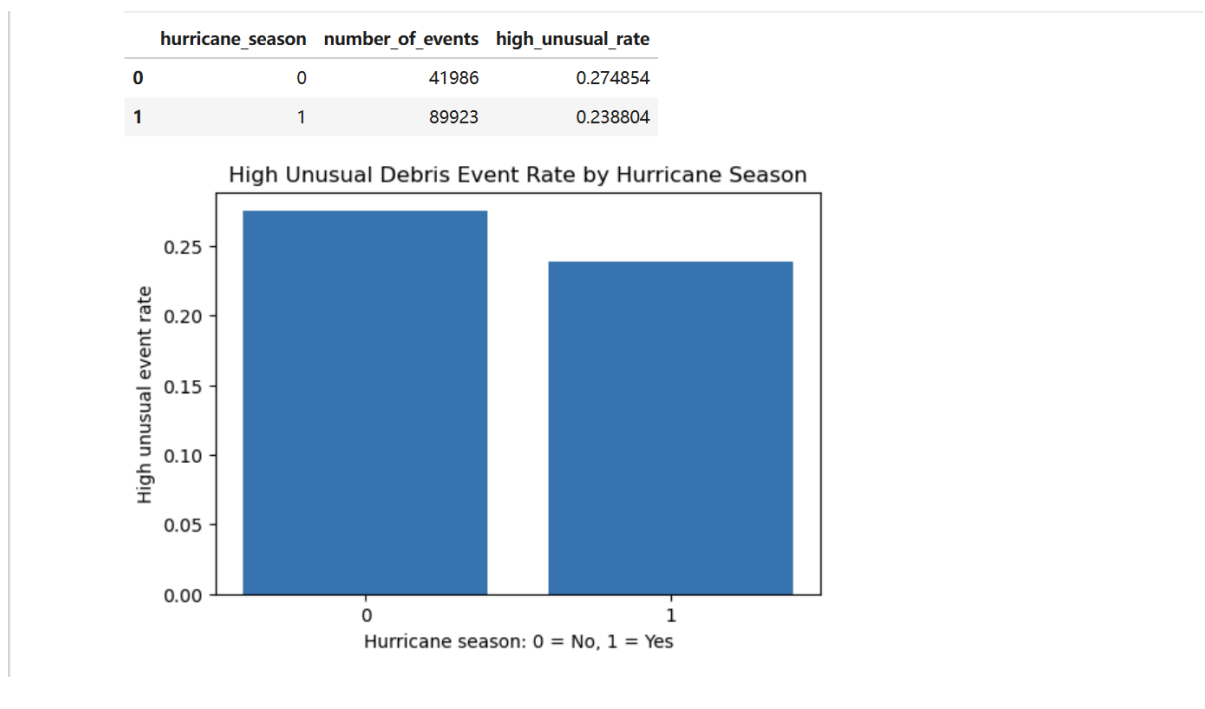


7.3 Hurricane Season Pattern

The hurricane-season comparison shows that high-unusual-debris events are not significantly higher during the hurricane season period in the full dataset. Non-hurricane-season cleanup events have a high-unusual rate of about 27.5%, while hurricane-season events have a slightly lower rate of about 23.9%.

However, it does not mean hurricanes have no effect on unusual debris. The variable `hurricane_season` only indicates whether a cleanup occurred between June and November. It does not measure whether the cleanup occurred after or before an actual hurricane, tropical storm, flooding event, or other special weather conditions. A more accurate storm-related analysis would require specific storm variables such as storm date, landfall location, rainfall, wind speed, etc.

Figure 3. High-unusual-debris event rate by hurricane season.



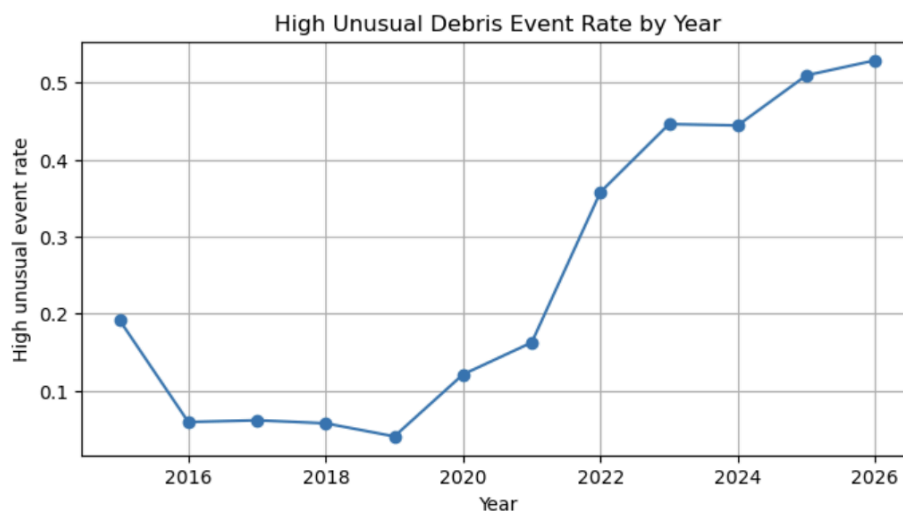
7.4 Yearly Pattern

The yearly pattern shows a strong upward pattern in high-unusual-debris event rate over time. From 2016 to 2019, the rate stayed relatively low, mostly around 4%–6%. Starting in 2020, the rate increased, and it rose sharply after 2021. By 2023 and 2024, more than 44% of cleanup events were labeled as high-unusual events, and the rate reached above 50% in 2025 and 2026.

This pattern indicates that **year** contains strong predictive information in the full dataset. However, the trend must be interpreted carefully. The data in 2015-2018 reports contain many cleanup events with “missing value” reports, and the report system is not convenient and sufficient as the current system, which may disturb the accuracy of data itself. This yearly pattern is the main reason for later building a separate 2023–2025 clean model.

Figure 4. High-unusual-debris event rate by year.

	year	number_of_events	high_unusual_rate
0	2015	878	0.191344
1	2016	5647	0.059324
2	2017	8032	0.061628
3	2018	9952	0.057677
4	2019	13781	0.040708
5	2020	14808	0.121151
6	2021	20384	0.162578
7	2022	14350	0.357700
8	2023	14415	0.445994
9	2024	14803	0.444302
10	2025	11428	0.509275
11	2026	3431	0.528709



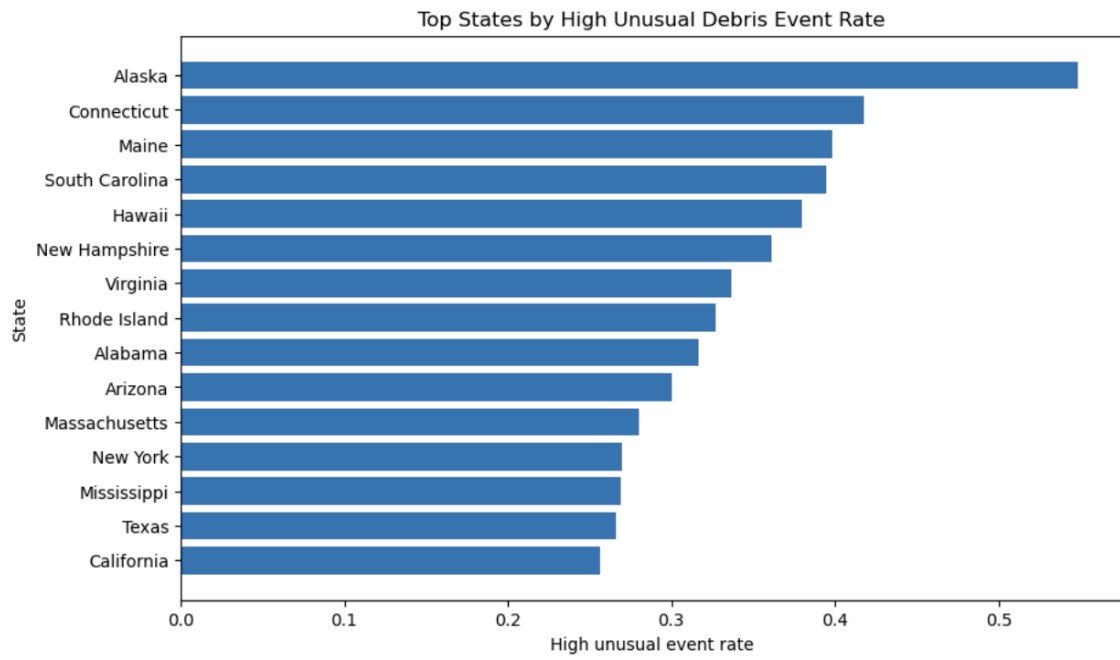
7.5 State-Level Pattern

The state-level analysis shows that high-unusual-debris event rates vary substantially across states. Among states with enough cleanup events to be included, Alaska has the highest high-unusual rate, followed by Connecticut, Maine, South Carolina, Hawaii, and New Hampshire. This suggests that high-unusual-debris events are geographically uneven rather than randomly distributed across the United States.

Still, state-level results should also be interpreted carefully. Some states have relatively small sample sizes, so their rates may be less stable. Large coastal states such as California, New York, and Florida have many more cleanup events, which makes their estimates more stable. Overall, the state-level analysis supports the idea that geographic variables should be included in the machine learning models.

Figure 5. Top states by high-unusual-debris event rate.

	state_name	number_of_events	high_unusual_rate
1	Alaska	463	0.548596
6	Connecticut	3883	0.417461
19	Maine	1075	0.398140
41	South Carolina	3083	0.394745
11	Hawaii	1831	0.379574
29	New Hampshire	567	0.361552
48	Virginia	4361	0.336620
40	Rhode Island	3828	0.327325
0	Alabama	1033	0.316554
2	Arizona	230	0.300000
21	Massachusetts	3125	0.280320
32	New York	6471	0.269819
24	Mississippi	349	0.269341
44	Texas	4426	0.266155
4	California	37322	0.256471
13	Illinois	1677	0.246869
49	Washington	3479	0.244898
7	Delaware	604	0.228477
38	Oregon	864	0.226852
35	Ohio	1837	0.221557

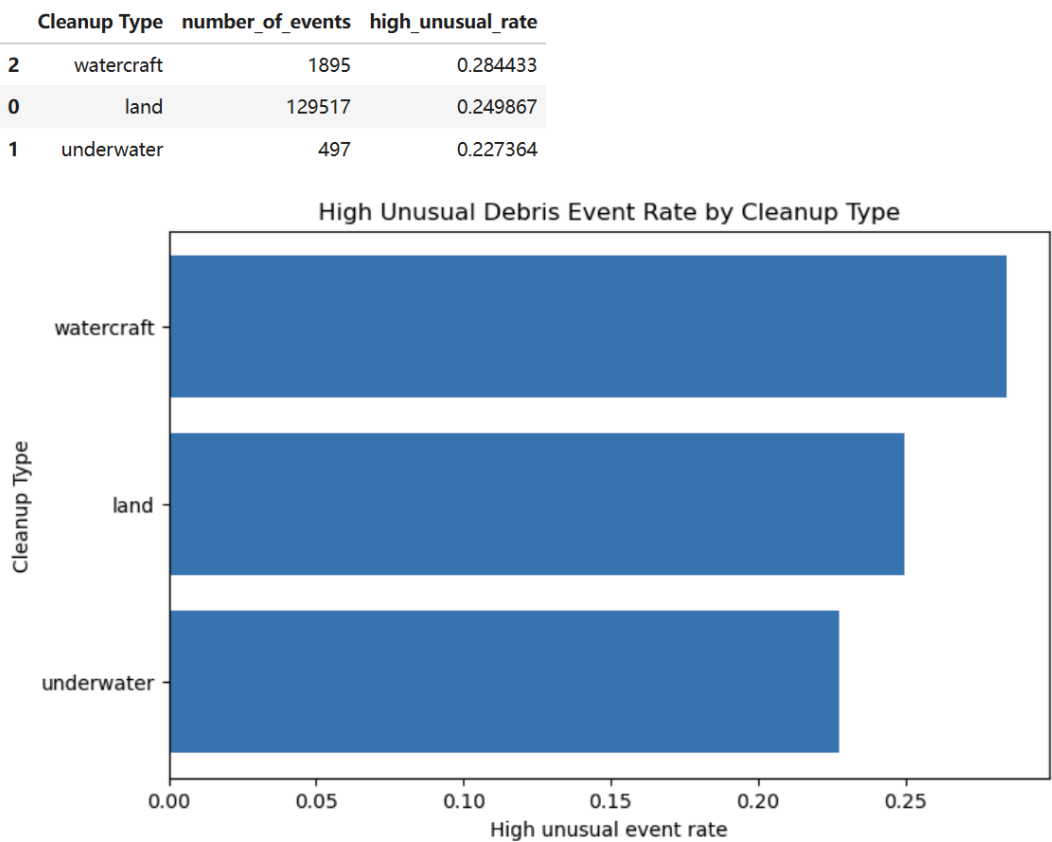


7.6 Cleanup Type Pattern

The cleanup-type analysis shows that watercraft cleanups have the highest high-unusual rate, followed by land cleanups and underwater cleanups. However, land cleanups dominate the dataset, while watercraft and underwater cleanups have much smaller sample sizes.

The differences across cleanup types are not very large. This suggests that cleanup type may provide some information but is probably less important than location. This is consistent with later feature-importance results, where **Cleanup Type** has low importance compared with geographic variables.

Figure 6. High-unusual-debris event rate by cleanup type.



8. Machine Learning Methodology

This project is a supervised binary classification task. The model uses cleanup-level features to predict whether `high_unusual_event = 1`.

The features used in the models include:

year, month, hurricane_season, Latitude, Longitude, People, Distance km, Area km, state_name, zone_name, Cleanup Type

Post-cleanup variables such as `Total Items Collected`, `unusual_debris_count`, and `unusual_share` were excluded from model features because they were used to define the target variable. Including them would create data leakage and make the model unrealistically strong.

Before modeling, numeric features were imputed using median values and standardized. Categorical features were imputed as "Unknown" and transformed using one-hot encoding. The data were split into training and test sets using stratification so that the class distribution remained approximately 75% normal events and 25% high-unusual events in both sets.

Three models were tested:

1. Logistic Regression: an interpretable linear baseline model.
2. Random Forest: a tree-based model that can capture nonlinear relationships and feature interactions.
3. Gradient Boosting: a flexible tree-based model that builds trees sequentially and often improves predictive performance.

Because the positive class represents only about 25% of observations, accuracy alone is not enough. I used precision, recall, F1-score, ROC-AUC, and PR-AUC to evaluate model performance.

9. Full 2015–2026 Model Results

9.1 Logistic Regression Baseline

Logistic Regression provides a baseline for the full 2015–2026 classification task. It achieves an accuracy of about 0.731, a ROC-AUC of about 0.803, and a PR-AUC of about 0.562. At the first glance, an accuracy of about 0.731 represents the 73.1% of the Logistic Regression model's prediction is correct, but since the model set 25% positive class rate, simply looking at accuracy and ROC-AUC is not enough. A PR-AUC of 0.562 is meaningfully above the random baseline, indicating the model is not random prediction but indeed finds some patterns .

For the positive class, Logistic Regression achieves a recall of about 0.748, meaning it identifies about 74.8% of true high-unusual-debris events. However, precision is about 0.476, meaning many predicted high-unusual events are false positives. Hence, this model is useful as an interpretable baseline, but it requires more flexible models to capture nonlinear patterns.

Figure 7. Logistic Regression Model Performance for the Full 2015–2026 Dataset.

```
Model: Logistic Regression
Accuracy: 0.7306117807596089
Precision: 0.47573189522342063
Recall: 0.7481447826745419
F1-score: 0.5816212397715901
ROC-AUC: 0.8028589417531388
PR-AUC: 0.5620714017928182

Confusion matrix:
[[14335  5444]
 [ 1663  4940]]

Classification report:
              precision    recall  f1-score   support

     0           0.90       0.72       0.80       19779
     1           0.48       0.75       0.58        6603

   accuracy                0.73       26382
  macro avg           0.69       0.74       0.69       26382
 weighted avg           0.79       0.73       0.75       26382
```

9.2 Random Forest

Random Forest improves slightly over Logistic Regression. Its accuracy increases to about 0.736, ROC-AUC increases to about 0.815, and PR-AUC increases to about 0.590. For the positive class, the recall improves to about 0.779. These slight improvements suggest that nonlinear relationships and feature interactions are useful for this prediction task. However, precision remains moderate at about 0.483, indicating that the model still produces a large number of false positives.

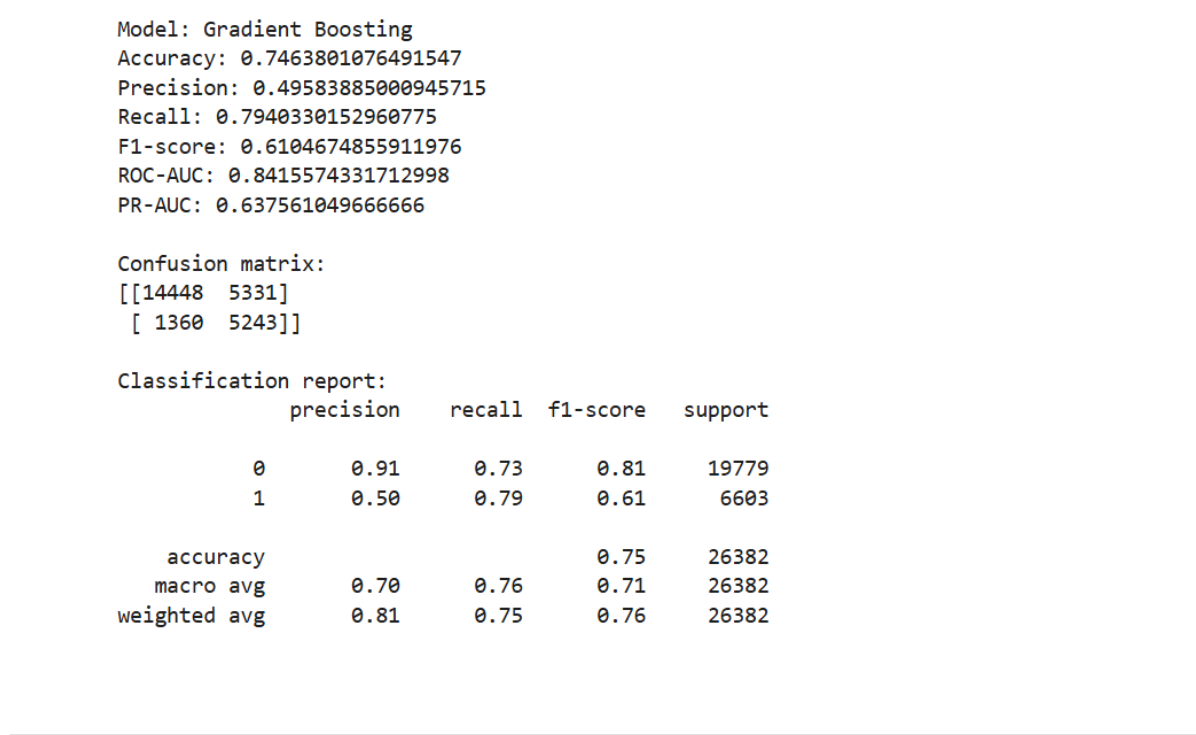
Figure 8. Random Forest Model Performance for the Full 2015–2026 Dataset.

Model: Random Forest				
Accuracy: 0.7363353801834585				
Precision: 0.4834069756510294				
Recall: 0.7787369377555656				
F1-score: 0.5965197215777263				
ROC-AUC: 0.8149657685316125				
PR-AUC: 0.5900263058686659				
Confusion matrix:				
[[14284 5495]				
[1461 5142]]				
Classification report:				
	precision	recall	f1-score	support
0	0.91	0.72	0.80	19779
1	0.48	0.78	0.60	6603
accuracy			0.74	26382
macro avg	0.70	0.75	0.70	26382
weighted avg	0.80	0.74	0.75	26382

9.3 Gradient Boosting

Gradient Boosting is the best-performing model in the full 2015–2026 analysis. It achieves the highest performance across all major metrics, including accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC. It achieves a ROC-AUC of about 0.842 and a PR-AUC of about 0.638. For the positive class, the Gradient Boosting model achieves a recall of about 0.794, meaning it identifies about 79.4% of true high-unusual-debris events. Its F1-score is about 0.610, which is higher than both the Logistic Regression model and the Random Forest model. However, the precision still remains as low as 0.496, implying a high possibility of false positives.

Figure 8. Gradient Boosting Model Performance for the Full 2015–2026 Dataset.



9.4 Model Comparison

The full 2015–2026 model comparison shows that Gradient Boosting performs best among the three models. Logistic Regression provides a strong baseline, with a ROC-AUC of about 0.803 and a PR-AUC of about 0.562. This suggests that even a linear model can identify meaningful predictive patterns in the full dataset.

Random Forest improves over the Logistic Regression on all major metrics. Its ROC-AUC increases to about 0.815, PR-AUC increases to about 0.590, and recall increases from about 0.748 to 0.779. This indicates that nonlinear relationships and feature interactions help predict high-unusual-debris events.

Gradient Boosting performs best overall. It achieves the highest accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC. Its ROC-AUC of about 0.842 suggests strong ability to distinguish high-unusual-debris events from normal events, while its PR-AUC of about 0.638 shows the best performance on the positive class.

Overall, the Gradient Boosting model is selected as the best full-sample model. However, this model should be used carefully. The later feature-importance analysis shows that **year** is the dominant predictor, requiring a robustness check of removing **year**. Besides, Gradient Boosting still has a relatively low precision of 0.496. Though the model has excellent performance in identifying the true high-unusual event, it still contains high probability of false positives.

Figure 9. Three Model Comparison for the Full 2015–2026 Dataset.

	model	accuracy	precision	recall	f1_score	roc_auc	pr_auc
0	Logistic Regression	0.730612	0.475732	0.748145	0.581621	0.802859	0.562071
1	Random Forest	0.736335	0.483407	0.778737	0.596520	0.814966	0.590026
2	Gradient Boosting	0.746380	0.495839	0.794033	0.610467	0.841557	0.637561

10. Full Model Feature Importance

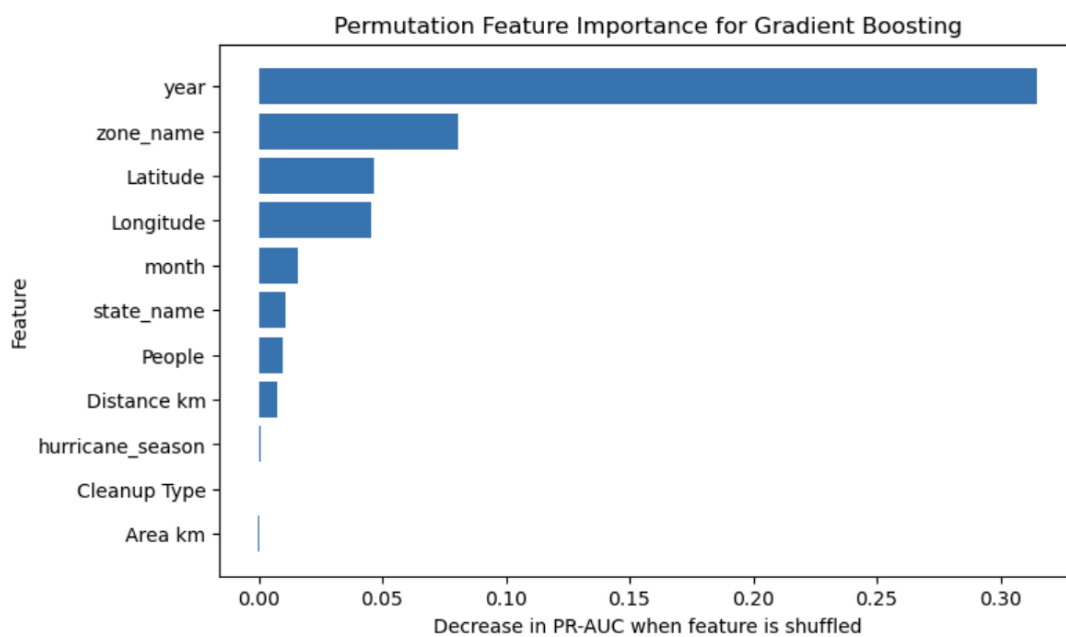
Permutation feature importance shows that **year** is by far the most important predictor in the full 2015–2026 Gradient Boosting model. When **year** is shuffled, the model’s PR-AUC decreases by about 0.315, which is much larger than the decrease caused by any other feature.

Following **year**, the next variables are **zone_name**, latitude, and longitude, suggesting that geography also contains meaningful predictive information. However, the dominance of **year** raises an interpretation concern. Because item category structure and reporting format changed across years, the model may be learning reporting or category-definition changes rather than only true environmental changes in debris patterns.

This result does not mean the full model is invalid, but it raises concern for the data sources. The data collected from 2015 to 2018 may have issues due to the limitation of the report system. Unintentional report errors may cause these years to have a lower high-unusual event rate that causes **year** to become an influential feature. On the other hand, the item categories of reports also have slight changes across years. As a conclusion, the model clearly finds predictive signals, but it requires further check.

Figure 10. Feature Importance for the Full 2015–2026 Dataset.

	feature	importance_mean	importance_std
0	year	0.314679	0.010350
9	zone_name	0.080452	0.006279
3	Latitude	0.046476	0.003285
4	Longitude	0.045769	0.005674
1	month	0.015851	0.001194
8	state_name	0.010988	0.001886
5	People	0.009746	0.001191
6	Distance km	0.007546	0.001425
2	hurricane_season	0.001010	0.000390
10	Cleanup Type	0.000384	0.000777
7	Area km	-0.000179	0.000063



11. Robustness Check: Removing Year

Because the full model relied heavily on `year`, I trained a no-year version of the Gradient Boosting model to test whether prediction performance would remain meaningful without this feature.

After removing `year`, model performance decreased substantially. ROC-AUC fell from about 0.842 to about 0.730, PR-AUC fell from about 0.638 to about 0.509, and F1-score fell from about 0.610 to about 0.494. This confirms that `year` contains substantial predictive information as a feature for the model using the 2015-2026 dataset.

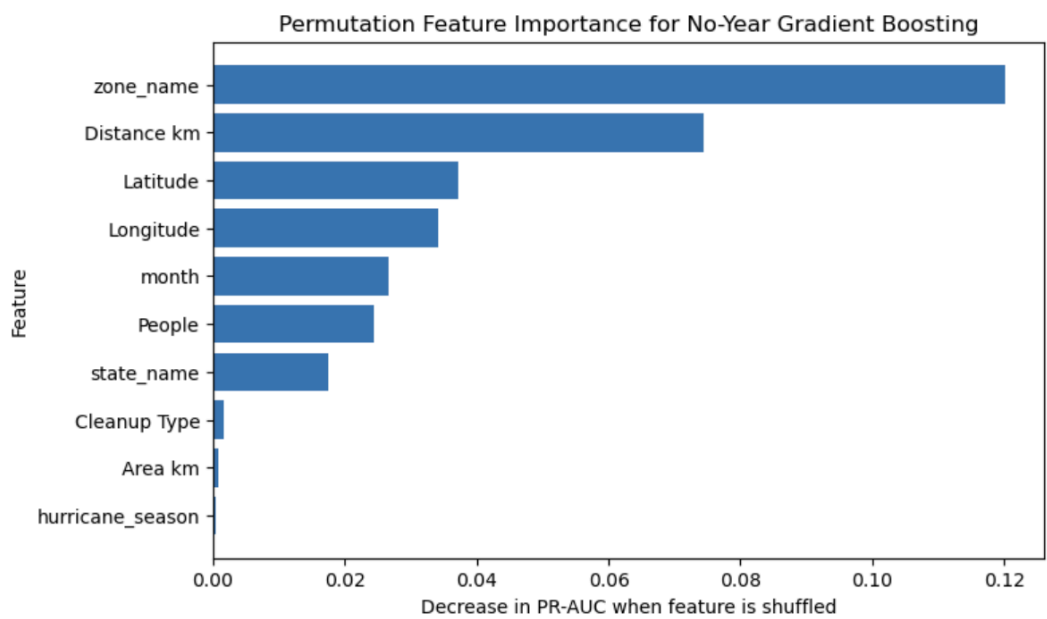
However, the no-year model still performs better than random prediction. Since the positive class rate is about 25%, a PR-AUC of about 0.509 is still meaningful. This suggests that other features, especially geographic and cleanup-level variables, also contain useful predictive signals beyond year. In the no-year model, the most important features are `zone_name`, `Distance km`, latitude, longitude, month, and number of people. This indicates that location and cleanup effort remain important even when the model cannot use year.

Figure 11. Robustness Check Removing Year for the Full 2015–2026 Dataset.

Model: Gradient Boosting No Year							
Accuracy: 0.6627245849442802							
Precision: 0.3955583871848548							
Recall: 0.6581856731788581							
F1-score: 0.4941444002274019							
ROC-AUC: 0.7298624892139775							
PR-AUC: 0.5094320892752181							
Confusion matrix:							
[[13138 6641]							
[2257 4346]]							
Classification report:							
		precision	recall	f1-score	support		
	0	0.85	0.66	0.75	19779		
	1	0.40	0.66	0.49	6603		
	accuracy			0.66	26382		
	macro avg	0.62	0.66	0.62	26382		
	weighted avg	0.74	0.66	0.68	26382		
		model	accuracy	precision	recall	f1_score	roc_auc
0		Gradient Boosting	0.746380	0.495839	0.794033	0.610467	0.841557
1		Gradient Boosting No Year	0.662725	0.395558	0.658186	0.494144	0.729862
							0.637561
							0.509432

Figure 12. Feature Importance Removing Year for the Full 2015–2026 Dataset.

	feature	importance_mean	importance_std
8	zone_name	0.120056	0.007880
5	Distance km	0.074351	0.006273
2	Latitude	0.037201	0.004785
3	Longitude	0.034168	0.003941
0	month	0.026569	0.003345
4	People	0.024361	0.000954
7	state_name	0.017554	0.001184
9	Cleanup Type	0.001574	0.000521
6	Area km	0.000728	0.000866
1	hurricane_season	0.000397	0.000213



12. Main Clean Model: 2023–2025

Because the full model was strongly influenced by `year`, I built a cleaner model using only 2023–2025 data. These years were selected because they have a more consistent item category structure and represent complete years. The report system using Apps on mobile phones or spreading on paper surveys from cleanup fields to collect the data. I exclude 2026 data from the clean model because 2026 is only a partial year. Besides, I still keep `year` as a feature in the clean model. Removing `year` would eliminate the possible influence of time changes on high-unusual events.

The 2023–2025 dataset was split into a training set and a test set. This process is similar to training the full year model. The training set contains 32,516 observations, and the test set contains 8,130 observations. The target distribution remained approximately 75% normal and 25% high-unusual in both sets, confirming that the stratified split worked correctly.

Figure 13. Target Distribution for 2023-2025 Clean Model.

```
Training set size:  
(32516, 11)
```

```
Test set size:  
(8130, 11)
```

```
Training target distribution:  
high_unusual_event_2325  
0      0.749508  
1      0.250492  
Name: proportion, dtype: float64
```

```
Test target distribution:  
high_unusual_event_2325  
0      0.749446  
1      0.250554  
Name: proportion, dtype: float64
```

13. 2023–2025 Clean Model Results

13.1 Logistic Regression

In the 2023–2025 clean dataset, Logistic Regression achieves an accuracy of about 0.668, a ROC-AUC of about 0.710, and a PR-AUC of about 0.465. This is lower than the full year Logistic Regression, but still meaningfully above the positive-class baseline.

For the positive class, the Logistic Regression model achieves a recall of about 0.597 and a precision of about 0.393. This means it identifies some high-unusual-debris events but also produces too many false positives. The result provides a reasonable baseline for the clean model.

Figure 14. Logistic Regression Model Performance for the Clean 2023-2025 Dataset.

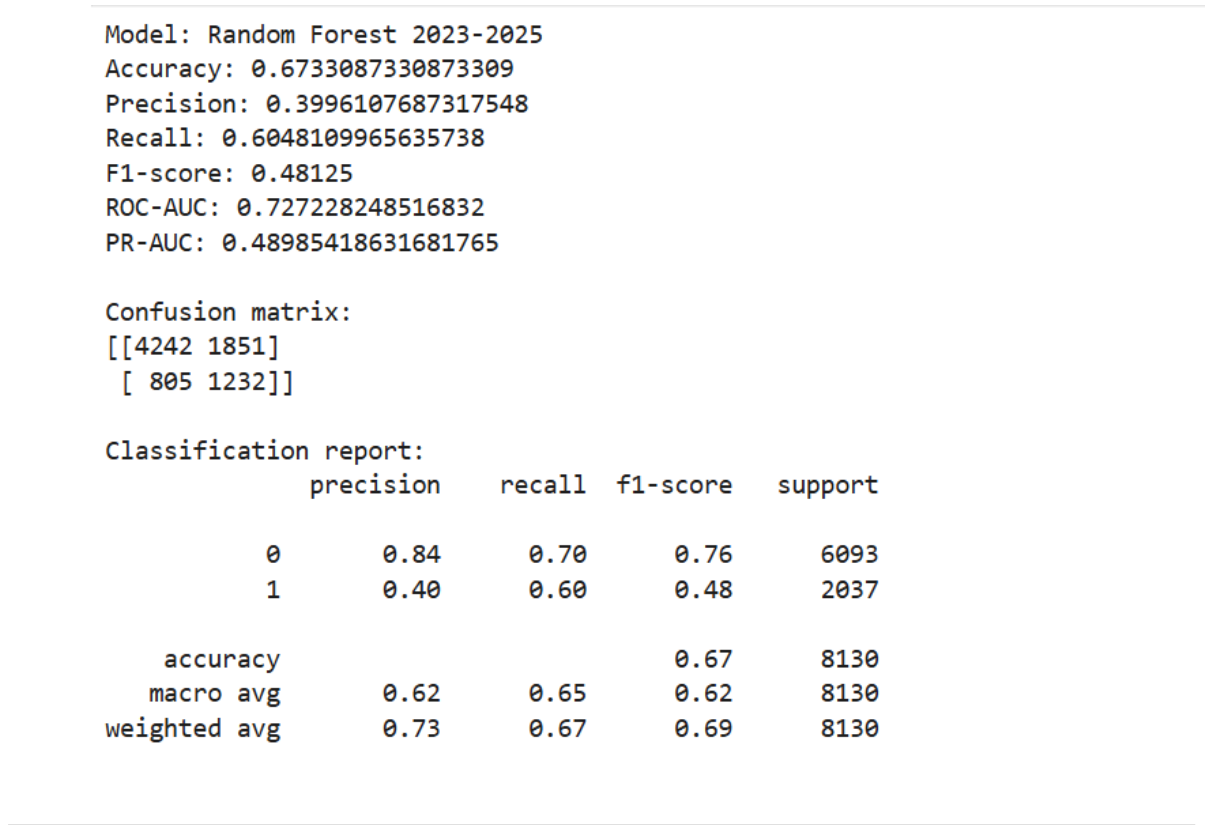
Model: Logistic Regression 2023-2025				
Accuracy: 0.6682656826568266				
Precision: 0.39327296248382926				
Recall: 0.5969563082965145				
F1-score: 0.47416650419185025				
ROC-AUC: 0.7100095387795825				
PR-AUC: 0.4652147563334771				
Confusion matrix:				
[[4217 1876]				
[821 1216]]				
Classification report:				
	precision	recall	f1-score	support
0	0.84	0.69	0.76	6093
1	0.39	0.60	0.47	2037
accuracy			0.67	8130
macro avg	0.62	0.64	0.62	8130
weighted avg	0.73	0.67	0.69	8130

13.2 Random Forest

Random Forest performs slightly better than Logistic Regression on the 2023–2025 clean dataset. It achieves an accuracy of about 0.673, a ROC-AUC of about 0.727, and a PR-AUC of about 0.490. Its recall for the positive class is about 0.605.

However, improvement over the Logistic Regression is modest, suggesting that nonlinear relationships help, but the Random Forest is still not the strongest model for this clean dataset.

Figure 15. Random Forest Model Performance for the Clean 2023-2025 Dataset.



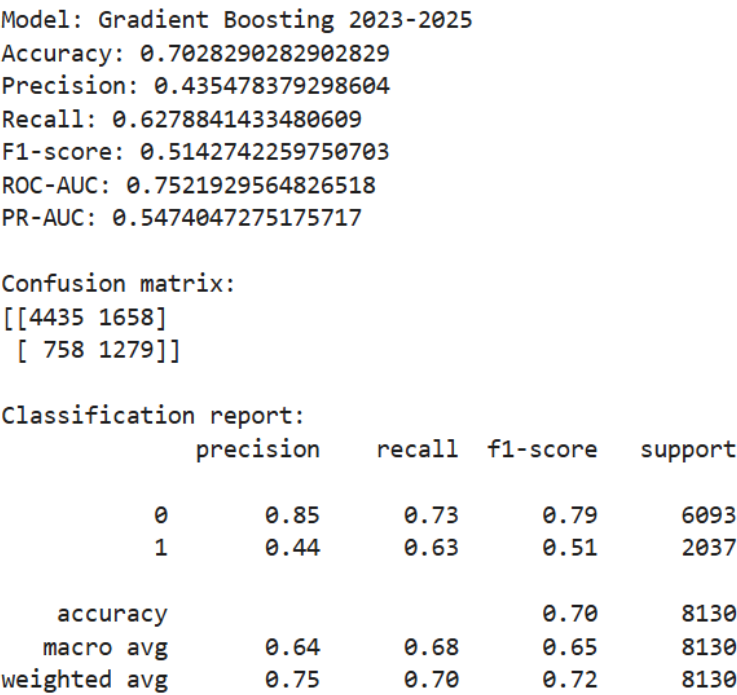
13.3 Gradient Boosting

Gradient Boosting is the best-performing model in the 2023–2025 clean dataset. It achieves the highest score across all major metrics, including accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC. The model achieves an accuracy of about 0.703, a ROC-AUC of about 0.752, and a PR-AUC of about 0.547.

For the positive class, Gradient Boosting achieves a recall of about 0.628, meaning it identifies about 62.8% of true high-unusual-debris events. Its F1-score is about 0.514, which is higher than the other two clean year models.

Although the 2023–2025 model performs lower than the full 2015–2026 model, it is more conservative and more interpretable because it uses more reliable data and avoids changes in item category structures.

Figure 16. Gradient Boosting Model Performance for the Clean 2023-2025 Dataset.



13.4 Model Comparison

The 2023–2025 model comparison shows that Gradient Boosting is the strongest model across all evaluation metrics. Logistic Regression provides a useful baseline, but its lower ROC-AUC and PR-AUC suggest that a linear model does not fully capture the patterns in the data. Random Forest improves slightly over the Logistic Regression, indicating that nonlinear relationships and feature interactions are useful. However, the improvement from Logistic Regression to Random Forest is relatively modest.

Gradient Boosting achieves a significant improvement. It has the highest accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC among the three models. Its ROC-AUC of about 0.752 indicates better overall separation between normal and high-unusual-debris events, while its PR-AUC of about 0.547 shows stronger performance on the positive class. Since high-unusual-debris events make up only about 25% of the dataset, PR-AUC is especially important.

As a conclusion, the Gradient Boosting model is considered as the best model for the 2023–2025 clean analysis. Although its performance is weaker than the full 2015–2026 model, it is more reliable for interpretation because it uses years with more reliable data and more consistent item category structures. The result suggests that the model can still identify meaningful high-unusual-debris patterns even after reducing the influence of cross-year reporting differences.

Figure 17. 3 Model Comparison for 2023-2025 Clean Model.

	model	accuracy	precision	recall	f1_score	roc_auc	pr_auc
0	Logistic Regression 2023-2025	0.668266	0.393273	0.596956	0.474167	0.710010	0.465215
1	Random Forest 2023-2025	0.673309	0.399611	0.604811	0.481250	0.727228	0.489854
2	Gradient Boosting 2023-2025	0.702829	0.435478	0.627884	0.514274	0.752193	0.547405

14. 2023–2025 Feature Importance

The feature-importance results for the 2023–2025 Gradient Boosting model show that geographic variables are the strongest predictors. The most important feature is `zone_name`, followed by longitude and latitude. This suggests that high-unusual-debris events are not evenly distributed across locations and that local cleanup zones contain strong predictive information.

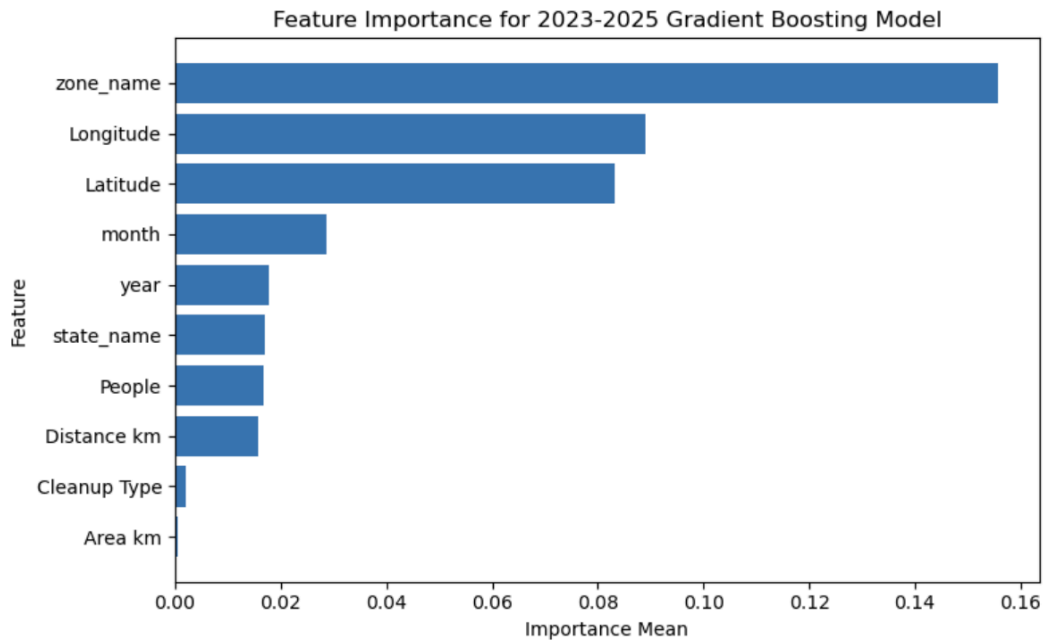
Compared with the full 2015–2026 model, `year` is much less dominant in the 2023–2025 model. In the full model, `year` is the most important feature and shows dominance compared with other features. In the clean model, `year` has a much smaller importance value, while location-related variables become the strongest predictors. This makes the 2023–2025 model more interpretable.

Months also have some predictive value, implying potential influences of seasonal changes. Cleanup effort variables such as `People` and `Distance km` have smaller importance but still positive. In contrast, `Cleanup Type`, `Area km`, and `hurricane_season` have very low importance.

The low importance of `hurricane_season` does not prove that storms are irrelevant. Instead, it suggests that a broad hurricane-season indicator is too simple to capture actual storm impacts. Future work should use more precise storm-event variables.

Figure 18. Feature Importance for the Clean 2023–2025 Dataset.

	feature	importance_mean	importance_std
9	zone_name	0.155649	0.005016
4	Longitude	0.089101	0.005089
3	Latitude	0.083084	0.001764
1	month	0.028575	0.004175
0	year	0.017662	0.002113
8	state_name	0.016892	0.004585
5	People	0.016822	0.000727
6	Distance km	0.015725	0.003537
10	Cleanup Type	0.001942	0.000574
7	Area km	0.000584	0.000448
2	hurricane_season	-0.000144	0.000189



15. Discussion

This project shows that machine learning can identify meaningful patterns in high-unusual-debris cleanup events. The full 2015–2026 model achieved the strongest performance, especially with Gradient Boosting. However, feature importance revealed that **year** dominated the full model, which created an important interpretation issue.

The no-year robustness check showed that model performance decreased when **year** was removed, confirming that year carried significant predictive information. However, the model still performed above random prediction, indicating that location and cleanup-level variables also contained meaningful signals.

The 2023–2025 clean model provides a more cautious and interpretable result. Although its performance is weaker than the full year model, its feature importance is more substantively meaningful. Instead of being dominated by **year**, the clean model relies most heavily on geography, especially **zone_name**, longitude, and latitude.

Overall, the results suggest that high-unusual-debris events are spatially uneven and associated with local cleanup condition. Broad indicators such as hurricane season and cleanup type are much less informative than more specific location variables.

16. Limitations

This project has several limitations:

- The definition of unusual debris is researcher-defined. Different researchers or organizations might classify unusual debris differently.
 - Item category structures changed across years. This makes the full 2015–2026 model difficult to interpret because the model may partly learn reporting changes rather than only environmental patterns.
 - The lack of a mature report system in 2015 to 2018 may cause the data in reports less reliable compared with recent years, causing potential bias in the final results.
 - 2026 is only a partial year, so it was excluded from the main clean model.
 - The `hurricane_season` variable is too broad. It does not indicate actual storm exposure, days since storm, storm severity, or whether a cleanup occurred after a hurricane.
 - Cleanup records may reflect volunteer activity, cleanup location choices, and reporting practices, not only true debris distribution.
 - The model identifies associations, not causal effects. The results should therefore be interpreted as predictive patterns rather than evidence that any specific feature causes high-unusual-debris events.
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17. Future Work

Future work could improve this project in several ways:

- Actual storm-event data should be added. This could include hurricane tracks, landfall dates, rainfall, wind speed, storm surge, and days since storm. These variables would provide a more accurate test of whether natural hazards are associated with unusual debris.
 - Future models could focus on a specific region, such as Florida or the Gulf Coast, where storm exposure and coastal cleanup activity are especially relevant. A county-level or beach-level model could also be used to create local unusual-debris risk maps.
 - Unsupervised learning methods such as clustering or anomaly detection could be used to identify unusual debris profiles without relying on a manually defined target variable.
 - A longer-term extension could frame cleanup planning as an optimization or reinforcement learning problem, where limited volunteer resources are allocated to locations with the highest expected unusual-debris risk.
-

18. Conclusion

This project demonstrates how machine learning can be used to predict cleanup events with unusually high shares of nonstandard coastal debris. The full 2015–2026 Gradient Boosting model achieved the strongest predictive performance, with a ROC-AUC of about 0.842 and a PR-AUC of about 0.638. However, its heavy reliance on `year` suggested that immature report systems in early years influence reliability of the data and changes in reporting structure or item category definitions may have influenced the results.

To produce a more interpretable model, I built a cleaner 2023–2025 model using years with more consistent category structures. In this clean model, Gradient Boosting again performed best, achieving

a ROC-AUC of about 0.752 and a PR-AUC of about 0.547. Feature importance showed that geographic variables, especially `zone_name`, longitude, and latitude, were the strongest predictors.

The main conclusion is that high-unusual-debris events can be predicted to some extent using cleanup-level data, but interpretation depends heavily on data consistency. The cleaner model suggests that unusual debris patterns are shaped more by geography and local cleanup context than by broad labels such as hurricane season or cleanup type. Future work should incorporate actual storm-event data to better connect unusual debris patterns with natural hazards and coastal resilience.